Drug Information & Pharmacology

Essential Medication Reference for Clinical Decision Support

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1. Analgesics & Anti-inflammatory Drugs

1.1 Non-Opioid Analgesics

Non-opioid analgesics are first-line agents for mild to moderate pain and form the cornerstone of the analgesic ladder. They lack the dependence potential of opioids but carry organ-specific toxicity risks.

Drug	Class	Usual Adult Dose	Key Cautions
Paracetamol (Acetaminophen)	Non-opioid	500–1000 mg q4–6h (Max: 4 g/day)	Hepatotoxicity in overdose or liver disease
Ibuprofen	NSAID	200–400 mg q6–8h (Max: 1200 mg/day OTC)	GI ulceration, renal impairment, cardiovascular risk
Naproxen	NSAID	250–500 mg q8–12h	Longer half-life; same NSAID risks
Aspirin	NSAID/Antiplatelet	75–300 mg/day (antiplatelet) 600 mg q4h (analgesic)	Reye syndrome in children; GI bleeding; anticoagulants
Celecoxib	COX-2 Inhibitor	100–200 mg BD	Cardiovascular risk; sulfonamide allergy

■ **NSAIDs are contraindicated in: active peptic ulcer disease, CKD (eGFR < 30), heart failure, third trimester pregnancy, and patients on anticoagulants without gastroprotection.**

1.2 Opioid Analgesics

Opioids act on mu, kappa, and delta receptors in the CNS and peripheral nervous system to produce analgesia. They are reserved for moderate-to-severe pain and must be prescribed with careful risk-benefit assessment.

- Codeine: Prodrug requiring CYP2D6 metabolism; avoid in ultra-rapid metabolisers and children
- Tramadol: Dual mechanism (mu-opioid + SNRI); lower addiction potential; lowers seizure threshold
- Morphine: Gold standard; titrate carefully; active metabolite accumulates in renal failure
- Oxycodone: High bioavailability; abuse-deterrent formulations available
- Fentanyl: Transdermal patch for chronic stable pain; do not use in opioid-naive patients

■ **EMERGENCY: Opioid overdose triad — miosis, respiratory depression, reduced consciousness. Naloxone 0.4 mg IV/IM/intranasal. Repeat every 2–3 min as needed. Monitor for re-narcotisation.**

2. Cardiovascular Medications

2.1 Antihypertensives

Modern hypertension management uses a stepwise approach combining agents from different classes to achieve target BP while minimising side effects. First-line options depend on comorbidities.

Drug Class	Examples	Mechanism	Preferred In
ACE Inhibitors	Enalapril, Ramipril, Lisinopril	Block ACE; reduce angiotensin II	Diabetes, CKD, post-MI, HFrEF
ARBs	Losartan, Valsartan, Telmisartan	Block AT1 receptor	ACE inhibitor-intolerant
CCBs	Amlodipine, Nifedipine	Block L-type Ca channels	Elderly, isolated systolic HTN
Beta-Blockers	Metoprolol, Bisoprolol, Atenolol	Block beta-1 adrenoceptors	IHD, HF, post-MI, arrhythmias
Thiazide Diuretics	Hydrochlorothiazide, Indapamide	Inhibit Na/Cl cotransporter	Isolated systolic HTN, elderly

■ ACE inhibitors and ARBs must not be combined (dual blockade increases risk of hyperkalaemia and acute kidney injury without additional cardiovascular benefit).

2.2 Antithrombotics

Antithrombotic drugs prevent formation or propagation of thrombi. They are classified into antiplatelets, anticoagulants, and fibrinolytics.

- Aspirin 75 mg: Irreversible COX-1 inhibition; used for secondary prevention of MI and stroke
- Clopidogrel: P2Y12 ADP receptor antagonist; combined with aspirin for ACS and stenting (DAPT)
- Warfarin: Vitamin K antagonist; narrow therapeutic index; multiple drug/food interactions; monitor INR
- Rivaroxaban / Apixaban / Edoxaban: Direct oral anticoagulants (DOACs); factor Xa inhibitors; no routine monitoring
- Dabigatran: Direct thrombin inhibitor; idarucizumab available as reversal agent
- Heparin UFH: Parenteral; monitor with aPTT; immediate onset for acute thrombosis

3. Antibiotics

3.1 Beta-Lactams

Beta-lactam antibiotics inhibit bacterial cell wall synthesis by binding to penicillin-binding proteins (PBPs). They include penicillins, cephalosporins, carbapenems, and monobactams.

Agent	Spectrum	Common Use	Key Adverse Effects
Amoxicillin	Gram +ve, some -ve	URTI, UTI, H. pylori	Rash, GI upset, diarrhoea
Amoxicillin-Clavulanate	Broad + beta-lactamase	Sinusitis, LRTI, skin infections	Cholestatic jaundice, diarrhoea
Flucloxacillin	Staphylococcal	Cellulitis, osteomyelitis (MSSA)	Hepatotoxicity (prolonged use)
Ceftriaxone (3rd gen)	Gram -ve, moderate +ve	Meningitis, sepsis, gonorrhoea	Biliary sludge, ceftriaxone-Ca precipitation
Meropenem	Ultra-broad	Multi-drug resistant organisms, sepsis	Seizures at high doses, nausea

■ Always document penicillin allergy history. Cross-reactivity with cephalosporins is ~1–2%. In true anaphylaxis to penicillin, use alternative class (e.g., azithromycin, clindamycin). Carry out allergy assessment before prescribing.

4. Psychiatric Medications

4.1 Antidepressants

Antidepressants are used for major depressive disorder, anxiety disorders, OCD, PTSD, and chronic pain syndromes. Onset of antidepressant effect typically requires 2–6 weeks.

- SSRIs (fluoxetine, sertraline, escitalopram): First-line; well tolerated; monitor for serotonin syndrome, QTc prolongation
- SNRIs (venlafaxine, duloxetine): Effective for depression and neuropathic pain; BP elevation at high doses
- TCAs (amitriptyline, clomipramine): Effective but cardiotoxic in overdose; anticholinergic effects
- MAOIs: Rarely used; serious tyramine and drug interactions; specialist initiation only
- Mirtazapine: Sedating; promotes appetite and sleep; useful in elderly or underweight patients

■ **Serotonin Syndrome: Triad of mental status change, autonomic instability, and neuromuscular abnormalities. Discontinue all serotonergic agents and seek emergency care for severe cases.**